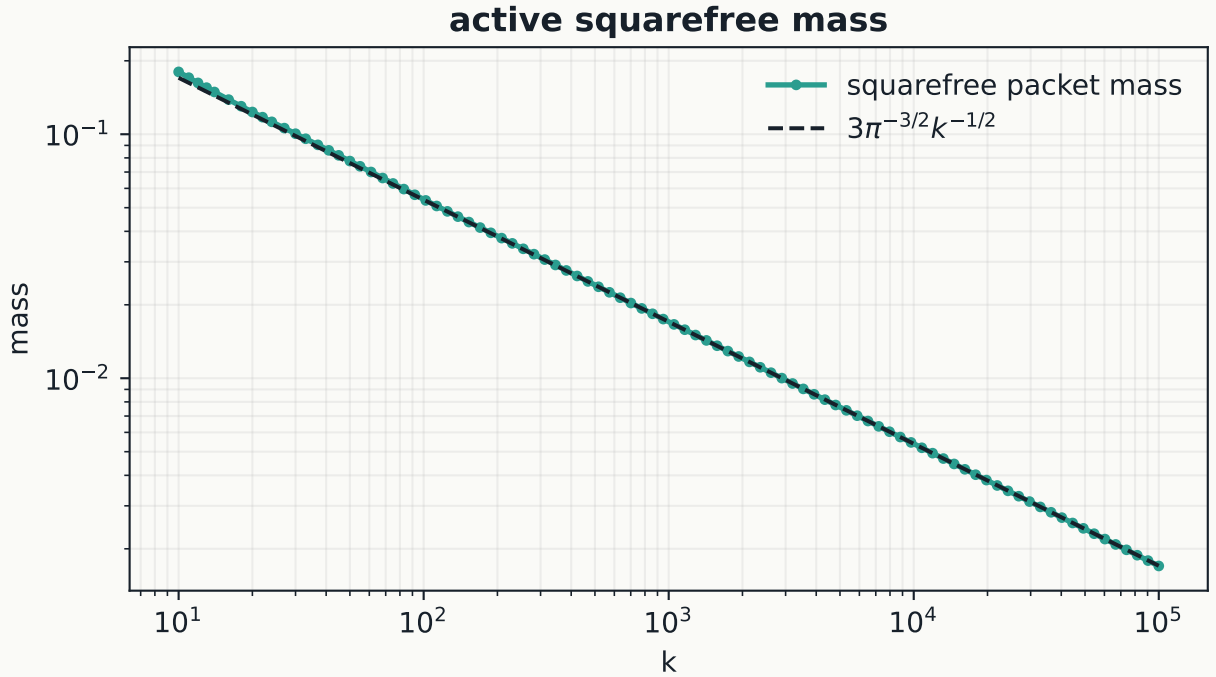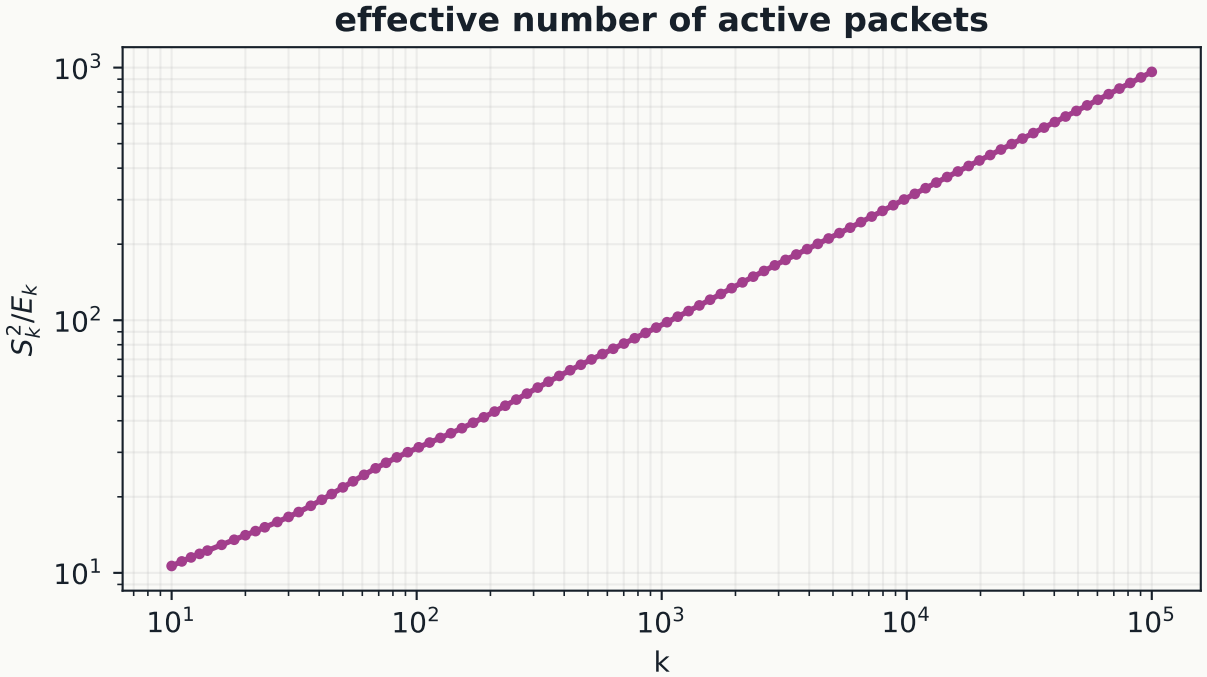The packet cloud grows like a constant multiple of $\sqrt{k}$